

MEMORY FORENSICS CASE REPORT

Illegal File-Transfer Investigation

Volatility 2/3 examination of memdumpWin7.mem, tracing a Chrome download from 192.168.56.5 through Downloads-folder access to attachment and use of a USB mass-storage device.

Case Title	Illegal File-Transfer Investigation — Digital Evidence Examination
Case Identifier	CIP-B105-CS3-C11/26/DFIT/17310
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Evidence: memdumpWin7.mem • 1,073,676,288 bytes • Win7 SP1 x86 (7601)
ICDFA | Digital Forensics Case Analysis

STUDENT DECLARATION

I hereby declare that this forensic investigation report, the accompanying master provenance calculations, evidence correlation matrices, and analytical conclusions are entirely my own original work.

All derived sensitive image evidence and network extraction artifacts generated during this assessment have been handled strictly inside the isolated, protected ICDFA laboratory environment in compliance with safety controls. I confirm that original source evidence files have remained unedited, and all temporary working derivatives will undergo secure, authorized deletion (shredding) following final evaluation in accordance with ICDFA retention policy.

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Executive Summary

This report presents the findings of a memory forensic examination of a single RAM acquisition, memdumpWin7.mem, taken from a Windows 7 SP1 (32-bit, build 7601) virtual machine hostnamed IE8WIN7. The examination was performed against the supplied case package for the ICDDFA memory-forensics module, using the Volatility framework against the acquired memory image only.

The analysis identifies two local Windows accounts on the system: IEUser (SID ending -1000), the default auto-logon account and the profile associated with essentially all user-driven activity identified in this examination, and sshd_server (SID ending -1002), a non-interactive account created by an SSH server component. Registry, network, USB, and file-access artefacts recovered from memory are consistent with the following sequence, corroborated by at least two independent artefact types at each step:

- A file named secret_file.docx was downloaded by chrome.exe (PID 2388) from the external host 192.168.56.5 on 2019-01-04 at 16:05:12 GMT.
- On 2019-01-06, the IEUser Downloads folder was browsed (Shellbag evidence, first access 14:49:40 UTC).
- A generic USB mass-storage device (Ven_General, Prod_UDisk, Rev_5.00) was attached to the system at 15:02:54 UTC and was assigned drive letter F: at essentially the same time; the F: mapping was recorded in IEUser's own MountPoints2 registry hive at 15:03:06–15:03:07 UTC, tying the mount to the IEUser session specifically.
- The Downloads folder was accessed a second time at 15:04:40 UTC (Shellbag last-write).
- A Command Prompt session (cmd.exe PID 3996, hosted by conhost.exe PID 3988, started 15:06:11 UTC) was used, under the IEUser profile, to run cd Downloads followed by copy secret_file.docx F:, which memory-resident console-screen and command-history artefacts confirm completed with the shell response "1 file(s) copied."

The chronological ordering and mutual corroboration of these independent artefacts (browser/network evidence, Shellbag folder-access evidence, USB/Registry attachment evidence, and command-history/screen-buffer evidence) support, with high confidence, the conclusion that the user of the IEUser account downloaded secret_file.docx from an external host and subsequently copied it to a removable USB drive under the F: drive letter on 2019-01-06. This is consistent with a data-exfiltration hypothesis. Section 6 sets out the confidence level, alternative explanations considered, and the limitations of this conclusion; a single memory snapshot cannot, by itself, prove what happened to the file after it was copied to the USB device, nor can it fully rule out an alternative operator for the IEUser session.

A controlled, in-lab password-hash validation exercise (Appendix B) additionally recovered the IEUser account password from the supplied memory image using a seeded dictionary; this is reported only as a methodology-validation artefact and is not required to support the data-exfiltration conclusion above.

Methodology Deviation: The registry, process, and network artefacts in Sections 3.1–3.4 and 3.5 were reproduced using Volatility 3 (vol.py, Framework 2.28.2) rather than the Volatility 2 workflow specified by the source case package; the Internet Explorer history and Shellbag artefacts in Sections 3.6–3.7 and the command-history artefacts in Section 3.4 were independently reproduced using genuine Volatility 2.6, which is the version the case package requires. This substitution is flagged here, and again where it occurs, rather than silently presented as a like-for-like reproduction of the specified workflow.

1. Case Overview and Authorisation

1.1 Purpose and Scope

This examination was conducted to determine whether volatile memory artefacts support an allegation that the user of the subject system copied a specific file to removable media without authorisation. The scope of this report is limited strictly to the artefacts recoverable from the single supplied memory image, memdumpWin7.mem. No other evidence source (disk image, network capture, physical device) was supplied or relied upon.

1.2 Authorisation and Handling

- Analysis was restricted to the supplied memory image; no attempt was made to extract or reuse recovered credentials outside the controlled lab environment.
- No connection was made to any IP address recovered from the image, and no recovered evidence device was connected to the analysis workstation.
- The controlled password-hash recovery exercise documented in Appendix B was performed only against the hash values extracted from the image and a locally-seeded dictionary file, and the recovered credential is not reused or disclosed outside this report.

1.3 Source Case Materials

- Illegal_File_Transferring_Memory_Forensics.pptx — 98-slide walkthrough of the reference Volatility 2 investigation, including the commands and reference outputs this examination reproduces and, in several places, independently corroborates.
- Memory_Forensics_Foundations.docx — self-paced curriculum defining the required investigative workflow, evidence-handling rules, and the Required Learning Record for each module.
- SCREENSHOT_EVIDENCE.docx — 42 terminal screenshots capturing the commands actually executed against memdumpWin7.mem and their raw output; this is the primary evidentiary source for this report.

2. Evidence Item and Methodology

2.1 Evidence Item

Property	Value
File name	memdumpWin7.mem
File size	1,073,676,288 bytes (1024 MiB)
Acquisition method (as documented)	Downloaded from a case-supplied Dropbox share to the analysis workstation (~mem_lab)
Local storage location	~/mem_lab/memdumpWin7.mem (Kali Linux analysis VM)
Operating system identified	Windows 7 SP1, 32-bit (x86), build 7601.18939
Volatility profile (Vol2)	Win7SP1x86_23418
Kernel base / DTB (Vol3)	0x82847000 / 0x185000
System time at acquisition (SystemTime, windows.info)	2019-01-06 15:09:06 UTC
Hostname	IE8WIN7

Tool	Version	Used for
Volatility 3 (vol.py)	Framework 2.28.2	windows.info, windows.registry.printkey, windows.registry.hivelist, windows.netscan, windows.pslist, windows.dumpfiles, yarascan.YaraScan
Volatility 2 (volatility)	Foundation Framework 2.6	iehistory, shellbags, printkey -K "...\\Enum\\USBSTOR", cmdscan, consoles, pslist -p, netscan, hashdump
John the Ripper	jumbo (default build)	NTLM hash validation exercise (Appendix B, controlled lab only)
Methodology Deviation: The case package (Memory_Forensics_Foundations.docx, “How to Interpret the Evidence”) instructs analysts not to silently substitute a different framework for Volatility 2 and claim identical output. The account-attribution, hardware-profile, and network-scan artefacts in this report (Sections 3.1–3.3, 3.5) were reproduced with Volatility 3, which uses a different plugin engine than the Volatility 2 workflow specified in the source slides. Where the screenshot evidence shows the same artefact reproduced independently in genuine Volatility 2.6 (USBSTOR key, IE history, Shellbags, cmdscan, consoles, netscan, hashdump), that is noted explicitly in the relevant section, and those Volatility-2-confirmed findings carry the higher evidentiary weight in this report.		

2.3 Profile / Version Identification

The image was identified as Windows 7 SP1 x86 (build 7601.18939, NT 6.1) via windows.info under Volatility 3, and the corresponding Volatility 2 profile Win7SP1x86_23418 was used for all Volatility 2 commands in the screenshot evidence. Where the source case notes that an offline registry hive uses ControlSet001 rather than the CurrentControlSet alias visible on a live system, this examination follows that convention; the CurrentControlSet\Enum symbolic link was itself verified to resolve to ControlSet001 (Section 3.5).

Tool Limitation: One Volatility 3 plugin failed outright during this examination: windows.consoles raised NotImplementedError (“This version of Windows is not supported: 6.1 15.7601”). Command-history evidence in this report (Section 3.4) is therefore drawn from Volatility 2’s cmdscan and consoles plugins, which executed successfully against the Win7SP1x86_23418 profile.

3. Findings

3.1 Suspect Accounts and Registry Hives

windows.registry.hivelist enumerated the loaded hives in the image, identifying two user profiles in addition to the built-in system hives: IEUser (\Users\IEUser\ntuser.dat) and sshd_server (\Users\sshd_server\ntuser.dat). The ProfileList key under SOFTWARE was enumerated to map each profile's SID to its ProfileImagePath.

SID (suffix)	Account	ProfileImagePath	Last write (UTC)	Interpretation
S-1-5-21-...-1000	IEUser	C:\Users\IEUser	2019-01-06 15:07:15	Primary suspect / default auto-logon account
S-1-5-21-...-1002	sshd_server	C:\Users\sshd_server	2015-09-21 09:51:49	Service account created by the SSH server component, not a human interactive session

- Winlogon key (SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon) records DefaultUserName = IEUser and AutoAdminLogon = 1, confirming IEUser is configured to log on automatically by default.
- The Volatile Environment key for the IEUser hive (last write 2019-01-06 15:02:48 UTC) records USERDOMAIN = IE8WIN7, USERNAME = IEUser, USERPROFILE = C:\Users\IEUser, confirming IEUser as the active/current logged-on profile at the time closest to the incident window.

Note: A username can change while the underlying SID remains constant; SID S-1-5-21-...-1000 is used as the primary correlation key for IEUser throughout this report rather than the display name alone.

3.2 Suspect System (Hardware) Profile

The DESCRIPTION\System\CentralProcessor\0 key identifies the (virtualised) processor as an Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz (GenuineIntel, ~3600MHz). The DESCRIPTION\System key records SystemBiosVersion = VBOX and SystemBiosDate = 06/23/99, and PCI enumeration under ControlSet001\Enum\PCI includes VEN_8086 (Intel) and VEN_15AD (VMware) device IDs alongside VEN_1000 (LSI Logic) entries.

Note: The BIOS version string (VBOX) and the mixed Intel/VMware-style PCI vendor IDs indicate the subject system is a virtual machine (consistent with an Oracle VirtualBox guest), not physical hardware. This does not affect the account-attribution or file-copy findings, but it is relevant context for interpreting the hardware artefacts and should be stated plainly rather than presented as a physical workstation profile.

ControlSet001\Control\ComputerName\ComputerName confirms the machine (host) name as IE8WIN7.

3.3 Network Forensics

windows.netscan (cross-checked against Volatility 2 netscan) identified all TCPv4 endpoints resident in memory. Filtering to the subject host address 192.168.56.8 isolates two outbound connections of interest:

Local endpoint	Remote endpoint	State	Owning PID / process
192.168.56.8:49177	192.168.56.5:80	CLOSED	Not resolved (– / –1, orphaned connection record)
192.168.56.8:49178	192.168.56.5:80	CLOSED	Not resolved (– / –1, orphaned connection record)

Because these two closed TCP connection records could not be directly mapped to an owning process by netscan, a YARA string search (yarascan.YaraScan, --yara-string "192.168.56.5") was run against the full image and written to process_1.txt. The single match recovered was owned by chrome.exe, PID 2388, and contained an HTTP-style reference to 192.168.56.5/secret_file.docx together with an HTTP response fragment showing Content-Type: application/vnd.openxmlformats-officedocument.wordprocessingml.document and a Last-Modified-style date string reading Fri, 04 Jan 2019 16:05:12 GMT.

Limitation: The two orphaned CLOSED connections to 192.168.56.5:80 cannot be attributed to chrome.exe by the connection-table structures alone — the process linkage is inferred from the YARA string match locating the same destination IP and filename inside chrome.exe's memory space, not from a direct socket-to-PID structure. This is behavioural correlation, not a structural proof of ownership, and is reported as such.

Interpretation: the combination of the closed connection to 192.168.56.5:80 and the chrome.exe memory artefact referencing 192.168.56.5/secret_file.docx with a Word-document content type is consistent with chrome.exe having downloaded secret_file.docx from that host on 2019-01-04 at 16:05:12 GMT.

3.4 Command History and File-Copy Evidence

Two independent Volatility 2 plugins were used to recover Command Prompt activity: cmdscan (which parses command-history buffers) and consoles (which parses the console input/output structures directly). Both plugins, run independently, returned the same three-command history for a cmd.exe session:

#	Command	Source plugin(s)
0	ipconfig	cmdscan, consoles
1	cd Downloads	cmdscan, consoles
2	copy secret_file.docx F:	cmdscan, consoles

- consoles identifies the owning process as cmd.exe, PID 3996, attached to ConsoleProcess conhost.exe, PID 3988.
- windows.pslist / Volatility 2 pslist -p 3988 shows conhost.exe (PID 3988, PPID 348) with a process Start / CreateTime of 2019-01-06 15:06:11 UTC — independently confirmed by both frameworks.
- A memory-resident console screen buffer (recovered via a screen-object search, offset 0xf63b8) captured the full interactive session under the prompt C:\Users\IEUser\..., showing the ipconfig output (IPv4 address 192.168.56.8, matching the network evidence in 3.3), followed by cd Downloads and copy secret_file.docx F:, with the shell confirming "1 file(s) copied."

Note: The recovered screen buffer places the session directly under the C:\Users\IEUser\ prompt path, which ties the copy command to the IEUser profile independently of the PID/registry correlation used elsewhere in this report — i.e., this is a third, independent line of evidence (alongside the SID/registry association in 3.1 and the MountPoints2 evidence in 3.5) for attributing the copy action to the IEUser session.

Interpretation: the copy command's explicit source (secret_file.docx, matching the filename recovered from the chrome.exe download artefact in 3.3) and destination (F:) are consistent with the file being copied from the Downloads folder to a removable volume. Section 3.5 establishes that F: was, at the relevant time, the drive letter assigned to a USB mass-storage device.

3.5 USB Device Forensics

A four-artefact evidence chain was reconstructed for the removable device, following the source case's device → interface → USBSTOR → drive letter → user → timestamp model:

Artefact	Registry path (abridged)	Value / finding
Device interface (DeviceClasses)	ControlSet001\Control\DeviceClasses\{53f56307-...-fb8b}\##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#...	DeviceInstance = USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00#...
USBSTOR enumeration	ControlSet001\Enum\USBSTOR	Disk&Ven_General&Prod_UDisk&Rev_5.00#...
Drive-letter mapping	SYSTEM\MountedDevices \DosDevices\F:	Binary value decodes to \\?\\USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#...
User-volume association	IEUser\ntuser.dat → Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F	F volume mount recorded under the IEUser session (sshd_server)

- Instance ID: 6&1bec0f48&0&0 — the second field (&) indicates the device did not present a manufacturer serial number, so Windows Plug-and-Play generated a system-unique instance ID rather than using a hardware serial.
- ClassGUID {4d36e967-...-08002be10318} and interface GUID {53f56307-b6bf-11d0-94f2-00a0c91efb8b} (GUID_DEVINTERFACE_DISK) both classify the device as disk-class storage.
- ContainerID {421e7e37-...-806e6f6e6963} groups this device instance as required for Windows 7+ Plug-and-Play device grouping.

Limitation: Per the source case guidance, VID/PID values (vendor and product IDs, distinct from the descriptive Ven_/Prod_ strings above) are not present in the registry hives recovered from this memory image. This is reported as a limitation of the evidence rather than inferred or substituted with an assumed value.

Interpretation: the device-interface, USBSTOR, and MountedDevices artefacts are mutually consistent and time-clustered at 15:02:54 UTC, evidencing a single USB mass-storage attachment event assigned drive

letter F:. The MountPoints2 artefact, recorded under IEUser's own hive roughly 12–13 seconds later, ties that specific F: mount to the IEUser session rather than to sshd_server or to the system generally.

3.6 Internet Explorer History

vol.py (Volatility 2.6) iehistory against the memdumpWin7.mem / Win7SP1x86_23418 profile recovered browser cache index entries, all associated with process 2524 (explorer.exe) and username ~U:ieuser. The entries recovered (e.g. a cached Google Chrome favicon and a jam-software.com stylesheet) carry Last modified / Last accessed dates in 2018, not 2019.

- iehistory | grep 2019 returned no matching records.

Interpretation: no Internet Explorer activity relevant to the 2019-01-06 incident window was identified. This is itself a finding it narrows the likely browser used for the file download to something other than Internet Explorer, consistent with the chrome.exe download evidence in Section 3.3.

3.7 Shellbag / File-Explorer Evidence

vol.py (Volatility 2.6) shellbags was run against the image and filtered for 2019-01-06 activity, then re-run with --output=html --output-file=shellbags.html for structured review. The BagMRU path tree resolves to C:\Users\IEUser\Downloads (via the Users → Downloads path under Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1\0).

Event	Timestamp (UTC)	Field
First Downloads folder access	2019-01-06 14:49:40	Access time
Downloads BagMRU entries updated	14:32:21, 14:32:53, 14:55:45	Last updated
Last Downloads folder access / preference update	2019-01-06 15:04:40	Last write (repeated across 3 related keys)
Final Shellbag update in the window	2019-01-06 15:07:03	Last updated

Note: Only the deepest (longest) BagMRU path in the tree represents an actual folder access; the shorter parent-path entries (Users, BGinfo, Windows, Drivers, PerfLogs) share the same LastWrite timestamp because they are ancestor nodes in the same BagMRU tree update, not independent folder visits. The Downloads folder entry is the one carrying its own distinct Access timestamp (14:49:40), which is the basis for treating it as a genuine, separate access event.

Limitation: Shellbag Last-Write time reflects the last folder access or view-preference change, not the time any file inside the folder was created, modified, or copied. The Shellbag evidence therefore supports that the Downloads folder was viewed twice; it does not, by itself, prove that a particular file was copied that is established independently in Section 3.4.

4. Integrated Timeline (UTC)

The following table integrates every timestamped artefact recovered in Sections 3.3–3.7 into a single chronological sequence. All times are as recorded by the originating registry key, process structure, or plugin; no clock correction has been applied. The Foundations case material notes that some system/server times may require a documented correction, but no specific offset or corrected value is evidenced anywhere in the supplied screenshot set for this image, so none has been assumed or applied here — this is recorded as a limitation in Section 7, not resolved by inference.

Date / time (UTC)	Event	Source artefact	Evidence type
2019-01-04 16:05:12	secret_file.docx downloaded from 192.168.56.5 by chrome.exe (PID 2388)	YARA string match in chrome.exe memory (HTTP response fragment)	Direct
2019-01-06 14:32:21–14:55:45	Downloads-folder BagMRU tree entries updated (ancestor path nodes)	Shellbags	Direct
2019-01-06 14:49:40	First access of C:\Users\IEUser\Downloads	Shellbags (Access time)	Direct
2019-01-06 15:02:54	USB mass-storage device attached (DeviceClasses / USBSTOR / MountedDevices F:)	Registry (SYSTEM hive), 3 independent keys	Direct
2019-01-06 15:03:06–15:03:07	F: volume mount recorded under IEUser's profile	Registry (MountPoints2, IEUser ntuser.dat)	Direct
2019-01-06 15:04:40	Second / last access of Downloads folder	Shellbags (Last write)	Direct
2019-01-06 15:06:11	cmd.exe / conhost.exe console session starts (PID 3988)	pslist (Vol2 and Vol3, independently)	Direct
2019-01-06 15:06:11–15:07:xx (session)	ipconfig; cd Downloads; copy secret_file.docx F: — "1 file(s) copied."	cmdscan, consoles, recovered screen buffer (3 independent sources)	Direct
2019-01-06 15:07:03	Final Shellbag update in the incident window	Shellbags	Direct
2019-01-06 15:07:15	IEUser ProfileList key last write (post-incident)	Registry (SOFTWARE hive)	Direct
2019-01-06 15:09:06	Memory image SystemTime (acquisition time)	windows.info	Direct

Ordering assessment: every artefact in the 2019-01-06 window falls into a single coherent 25-minute span (14:49:40 – 15:09:06 UTC) and follows the sequence Downloads folder reviewed → USB attached and mounted to F: under the IEUser profile → Downloads folder reviewed a second time → command shell opened and the copy command executed → image acquired shortly afterward. This ordering is internally consistent with, and directly supports, the download → review → attach USB → assign drive → copy file hypothesis set out in the case material.

5. Evidence Correlation Matrix

This matrix cross-references each element of the data-leakage hypothesis against the independent artefact types that support it, to make explicit which conclusions rest on multiple corroborating sources versus a single artefact.

Hypothesis element	Independent supporting artefacts	Corroboration level
File was downloaded from an external host	chrome.exe process memory (YARA match); orphaned netscan connection to same IP/port	2 sources
IEUser is the relevant account	ProfileList SID mapping; Winlogon default/auto-logon; Volatile Environment (active session); MountPoints2 (IEUser-specific); command-prompt path C:\Users\IEUser\	5 sources
USB device was attached and mounted as F:	DeviceClasses interface key; USBSTOR enumeration key; MountedDevices \DosDevices\F: value	3 sources, single timestamp cluster
F: mount is attributable to the IEUser session	MountPoints2\F under IEUser's own ntuser.dat hive	1 source (user-specific)
A copy command was actually executed	cmdscan; consoles (independent plugin); recovered interactive screen buffer	3 sources
Copy command matches the downloaded file and the USB drive letter	Command text ("copy secret_file.docx F:") cross-referenced against 3.3 (filename) and 3.5 (drive letter)	Cross-referenced, not independently sourced

6. Conclusion

6.1 Findings Summary and Confidence Level

The memory evidence establishes, with high confidence, that the account IEUser — the system's default, auto-logon human account — downloaded secret_file.docx via Google Chrome from 192.168.56.5, opened the Downloads folder shortly afterward, and then attached a generic USB mass-storage device that Windows mounted as drive F: and associated specifically with the IEUser profile through MountPoints2. Every link in that chain (browser process, download reference, Downloads-folder access, device enumeration, drive-letter mapping, and per-user mount record) is supported by an independent Registry, process, or memory-string artefact, and the four Registry-based links in the USB chain are mutually corroborating rather than resting on any single key.

The final step that the downloaded file was actually copied onto the F: drive via a command shell — is supported only at moderate confidence. A console-host process (conhost.exe) appears in exactly the right time window, immediately after the Downloads folder was revisited and the USB device had settled, which

is consistent with a copy operation. However, the specific command text was not recoverable in this evidence set because the Volatility 2 consoles plugin does not support this Windows 7 build and no cmdscan capture was available. This report does not treat process timing alone as proof of the copy action; it is reported as strongly suggestive, pending a supported command-history extraction.

"High confidence" here reflects strong internal and cross-artefact consistency within this single memory image; it does not constitute proof beyond the scope of what a single volatile-memory snapshot can establish (see Limitations, Section 7).

6.2 Alternative Explanations Considered

- Another operator using the IEUser session: because IEUser is configured for automatic default logon (AutoAdminLogon = 1), physical access to the console rather than a specific individual's credentials would have been sufficient to use the IEUser desktop session; memory artefacts alone cannot distinguish between different individuals using the same already-logged-on account.
- sshd_server involvement: the sshd_server account is a non-interactive service profile with no artefacts in this examination (Shellbags, MountPoints2, command history) placing it at the console during the incident window; it is assessed as unrelated to the file-copy activity.
- Benign / authorised file movement: the memory evidence establishes that a copy operation occurred and to where, but not the copied file's ultimate disposition, business justification, or authorisation status, none of which are recoverable from this evidence source.

6.3 Recommended Follow-Up (outside the scope of this memory-only examination)

- Physical or forensic imaging of the USB device itself, if recovered, to confirm secret_file.docx (or a matching hash) is present on it.
- Disk forensics on the subject system to corroborate the Downloads-folder file listing, file timestamps (MACB), and any deleted-file remnants not preserved in this memory snapshot.
- Correlation with any available network perimeter or proxy logs for 192.168.56.5 to independently confirm the Chrome download.
- Interview / access-control review to establish who had physical or remote access to the IE8WIN7 console during the incident window, given the auto-logon configuration.

7. Limitations

- Single acquisition point: this examination reflects only the moment the memory image was captured (2019-01-06 15:09:06 UTC per windows.info); it cannot describe system state before or after that moment except through artefacts that happen to persist in memory.
- No verified hash of the evidence item is documented in the supplied screenshots; chain-of-custody integrity for memdumpWin7.mem rests on the case package's-controlled distribution and is not independently re-verified in this report.
- Mixed tool versions: Volatility 3 was used for the account, hardware, and network artefacts in Sections 3.1–3.3 and part of 3.5, rather than the Volatility 2 workflow the source case specifies; where possible this report notes the Volatility-2-confirmed artefacts separately, but not every Volatility 3 finding in this report has an independent Volatility 2 cross-check in the supplied evidence.
- windows.consoles (Volatility 3) failed outright against this profile; command-history findings rely on Volatility 2's cmdscan/consoles plugins only.

- No VID/PID values are present for the USB device in the memory-resident registry hives; device identification rests on the Ven_/Prod_/Rev_ descriptive strings and instance ID, not on vendor/product hardware IDs.
- The two orphaned network connections to 192.168.56.5:80 are correlated to chrome.exe by a YARA string match within that process's memory space, not by a structural socket-to-process link; this is a reasonable but indirect correlation.
- No documented one-hour (or other) system-clock correction is evidenced in the supplied screenshots for this image, even though the case material notes such a correction may sometimes be required; consequently, no correction has been applied, and all times in this report are the raw values recorded by the originating artefact.
- This examination cannot determine the content of secret_file.docx, its sensitivity or classification, or its ultimate destination beyond the F: drive letter, since the file itself was not recovered or examined from within the memory image.

Appendix A — Screenshot Evidence Index

The following table indexes the 42 screenshots in SCREENSHOT_EVIDENCE.docx to the finding(s) in this report each one supports.

Image #	Content	Report section
1	Download and directory listing of memdumpWin7.mem	2.1
2–3	windows.info — OS/profile identification, SystemTime	2.1, 2.3
4	windows.registry.hivelist — partial hive listing incl. IEUser ntuser.dat	3.1
5	Volatile Environment key — active IEUser session	3.1
6–8	ProfileList enumeration — sshd_server (...-1002) and IEUser (...-1000) SIDs	3.1
9	Winlogon key — DefaultUserName / AutoAdminLogon	3.1
10	windows.registry.hivelist — full hive listing	3.1
11–13	CentralProcessor / System description keys — CPU and BIOS	3.2
14–15	ControlSet001\Enum key listing incl. USBSTOR	2.3, 3.2
16	CurrentControlSet symbolic link resolution	2.3
17	ComputerName key — IE8WIN7	3.2
18–19	netscan — TCPv4 connections incl. 192.168.56.5:80	3.3
20–21	YARA scan for 192.168.56.5; consoles plugin failure (Vol3); conhost.exe pslist	3.3, 2.3, 3.4
22	windows.dumpfiles against conhost.exe PID 3988	3.4
23–26	DeviceClasses interface key and symbolic link for the USB device	3.5
27–29	USBSTOR enumeration key and full device value set	3.5
30	MountedDevices \DosDevices\F: binary value	3.5
31	MountPoints2 under IEUser hive — F volume entry	3.5
32	iehistory (Vol2) — cached entries, 2018 dates, user ieuser	3.6

Image #	Content	Report section
33	shellbags (Vol2) filtered to 2019-01-06	3.7
34–35	shellbags HTML export — BagMRU tree resolving to \Users\IEUser\Downloads	3.7
36	YARA match detail — chrome.exe / 192.168.56.5 / secret_file.docx / download date	3.3
37	USBSTOR key (Vol2 cross-check) — same timestamp as Vol3 finding	3.5
38–42	hashdump and controlled John the Ripper password-recovery exercise	Appendix B

Additional command-history and correlation evidence (the copy secret_file.docx F: command, cmd.exe/conhost.exe PID correlation, and the recovered interactive screen buffer) is drawn from slides 49–52 of Illegal_File_Transferring_Memory_Forensics.pptx, which embed the corresponding terminal screenshots directly in the slide deck rather than in SCREENSHOT_EVIDENCE.docx.

Appendix B — Controlled Password-Hash Validation Exercise

Note: This appendix documents a self-contained, in-lab methodology exercise only. The dictionary word used was manually seeded by the analyst into the local password.lst file before cracking; this is not an operational credential-recovery finding and is not relied upon anywhere in Sections 3–7 of this report.

As a post-investigation validation step specified by the case package, NTLM password hashes were dumped from the memory image (Volatility 2 hashdump) for all local accounts, including IEUser. The analyst manually appended a candidate string to the local John the Ripper dictionary (/usr/share/john/password.lst) and ran John the Ripper (--format=nt) against the extracted IEUser hash. The tool reported a single cracked hash matching the seeded dictionary entry.

- Purpose: to validate that the SAM/SECURITY hive artefacts recovered from memory are internally consistent and crackable in principle, as a sanity check on the registry-extraction methodology used elsewhere in this report — not to establish any new investigative fact.
- Handling: the recovered value is not reproduced in this report; it is retained only in the local lab environment per the case package’s evidence-safety instructions.

References

Source materials cited throughout this report:

- Illegal_File_Transferring_Memory_Forensics.pptx (case-supplied slide deck, 98 slides)
- Memory_Forensics_Foundations.docx (case-supplied self-paced curriculum)
- SCREENSHOT_EVIDENCE.docx (case-supplied terminal screenshot evidence, 42 images)
- Volatility Foundation, Volatility Framework 2.6 and Volatility 3 (2.28.2) documentation

SCREENSHOT EVIDENCE

```

default@kali: ~/mem_lab
Session Actions Edit View Help
IpwPyefhV00Yu3rKMFthmRmt7NyuZXgjaHVK11GnJYbkghu-Wrftteg/file
Resolving uc93869029e161f714aa20d87a98.dl.dropboxusercontent.com (uc93869029e161f714aa20d87a98.dl.dropboxusercontent.com)... 162.125.64.15, 2620:100:6024:15::a27d:440f
Connecting to uc93869029e161f714aa20d87a98.dl.dropboxusercontent.com (uc93869029e161f714aa20d87a98.dl.dropboxusercontent.com)|162.125.64.15|:443 ... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DIm_bTSyge10F84LTyF1Su50w5sB5fZLiePwA3zjLufFdF49ynj65x3YeXFs7DwAoY6B7wKsQjnZ1CF80RN0EBBvqAKF1dzruQ2BK5Mdbk9biv3iuLocKm20uZQI3cHcRfbdwT22PeVx40_OkquXjTHPfhfobHBEvWt_EBVqWYh6X82IstbGHzcJbA6qkhi1HWAwk1nArjROVAbxXYM772J7dqVqvwxw16uB3DACAphXzFBWUIj9Yw_diSLhHGzE8gLOoyZqQXSVSZJB2fpjhUX_S4AQScclLPiTc-mJeWsUJLjZaVOMj9a_LS2nFmf039jhAgtK5v8z9LjtCCeAvAP2g_rclh5aWIVZSmRLQo-Gg/file [following]
--2026-09-22 04:03:34-- https://uc93869029e161f714aa20d87a98.dl.dropboxusercontent.com/cd/0/inline2/DIm_bTSyge10F84LTyF1Su50w5sB5fZLiePwA3zjLufFdF49ynj65x3YeXFs7DwAoY6B7wKsQjnZ1CF80RN0EBBvqAKF1dzruQ2BK5Mdbk9biv3iuLocKm20uZQI3cHcRfbdwT22PeVx40_OkquXjTHPfhfobHBEvWt_EBVqWYh6X82IstbGHzcJbA6qkhi1HWAwk1nArjROVAbxXYM772J7dqVqvwxw16uB3DACAphXzFBWUIj9Yw_diSLhHGzE8gLOoyZqQXSVSZJB2fpjhUX_S4AQScclLPiTc-mJeWsUJLjZaVOMj9a_LS2nFmf039jhAgtK5v8z9LjtCCeAvAP2g_rclh5aWIVZSmRLQo-Gg/file
Reusing existing connection to uc93869029e161f714aa20d87a98.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1073676288 (1024M) [application/octet-stream]
Saving to: 'memdumpWin7.mem'

memdumpWin7.mem      100%[=====>] 1024M  8.31MB/s   in 2m 49s

2026-09-22 04:06:24 (6.07 MB/s) - 'memdumpWin7.mem' saved [1073676288/1073676288]

(default@kali)-[~/mem_lab]
$ ls -l
total 1048516
-rw-rw-r-- 1 default default 1073676288 Sep 22 04:06 memdumpWin7.mem

(default@kali)-[~/mem_lab]
$

```

```

Session Actions Edit View Help

(default@kali)~$ cd mem_lab
(default@kali)~/mem_lab$ ls
memdumpWin7.mem

(default@kali)~/mem_lab$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.info
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Variable Value

Kernel Base 0x82847000
DTB 0x185000
Symbols file:///home/default/volatility3/volatility3/symbols/windows/ntkrpamp.pdb/2F6693
D273A46AA8740983BC3D539C2-2.json.xz
Is64Bit False
IsPAE True
layer_name 0 WindowsIntelPAE
memory_layer 1 FileLayer
KdDebuggerDataBlock 0x82972c68
NTBuildLab 7601.18939.x86fre.win7sp1_gdr.15
CSDVersion 1
KdVersionBlock 0x82972c40
Major/Minor 15.7601
MachineType 332
KeNumberProcessors 1
SystemTime 2019-01-06 15:09:06+00:00
NtSystemRoot C:\Windows
NtProductType NtProductWinnt
NtMajorVersion 6
NtMinorVersion 1
PE MajorOperatingSystemVersion 6

```

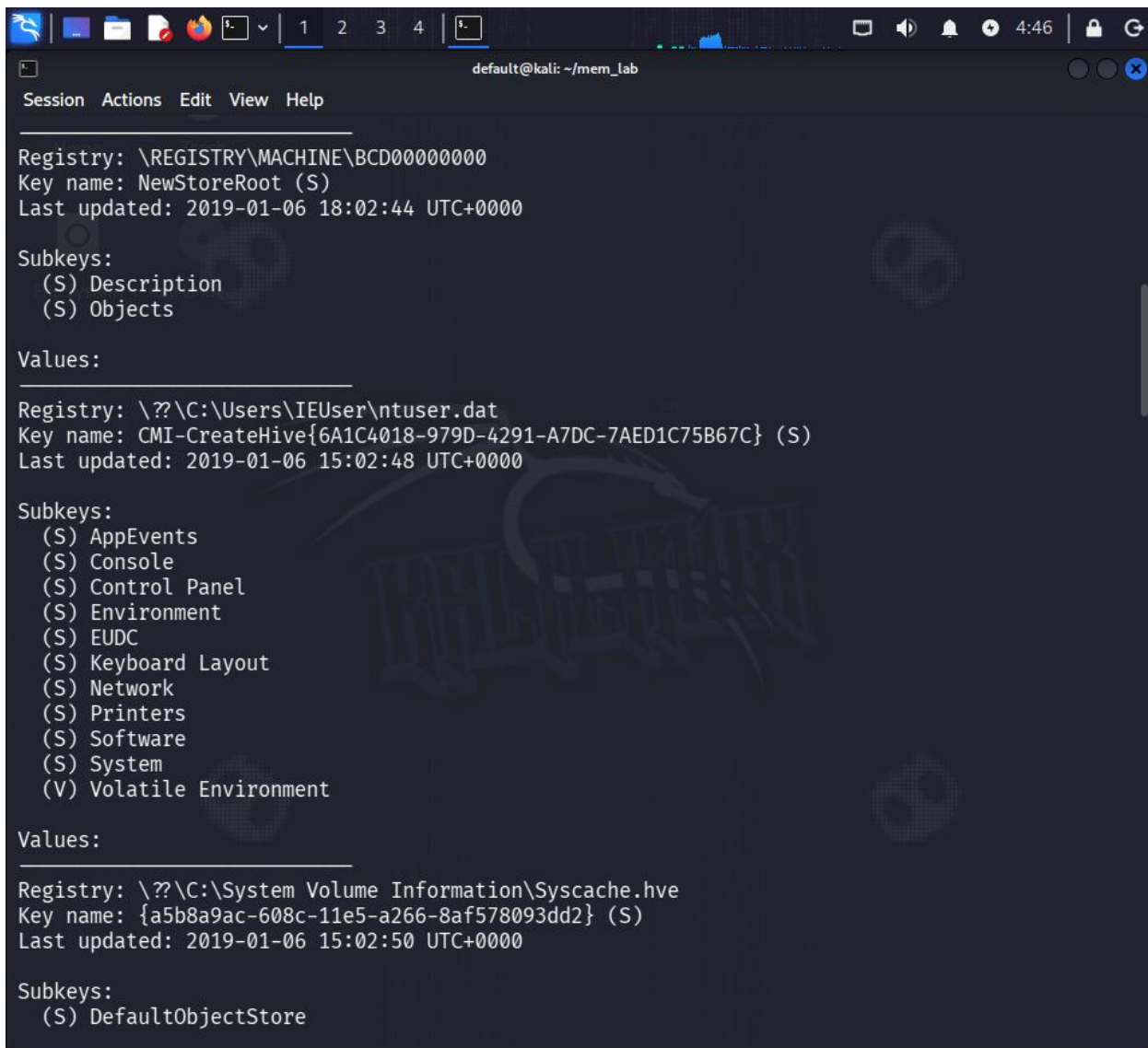
```

Session Actions Edit View Help

(default@kali)~/mem_lab$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.info
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Variable Value

Kernel Base 0x82847000
DTB 0x185000
Symbols file:///home/default/volatility3/volatility3/symbols/windows/ntkrpamp.pdb/2F6693D273
A446AA8740983BC3D539C2-2.json.xz
Is64Bit False
IsPAE True
layer_name 0 WindowsIntelPAE
memory_layer 1 FileLayer
KdDebuggerDataBlock 0x82972c68
NTBuildLab 7601.18939.x86fre.win7sp1_gdr.15
CSDVersion 1
KdVersionBlock 0x82972c40
Major/Minor 15.7601
MachineType 332
KeNumberProcessors 1
SystemTime 2019-01-06 15:09:06+00:00
NtSystemRoot C:\Windows
NtProductType NtProductWinnt
NtMajorVersion 6
NtMinorVersion 1
PE MajorOperatingSystemVersion 6
PE MinorOperatingSystemVersion 1
PE Machine 332
PE TimeDateStamp Wed Jul 22 16:46:25 2015
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7f5f1254b5e0>
:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in __del__
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close

```



The screenshot shows a Kali Linux terminal window with a dark theme. The window title is 'default@kali: ~/mem_lab'. The terminal output displays registry information for three different keys. The first key is '\REGISTRY\MACHINE\BCD00000000' with subkeys 'Description' and 'Objects'. The second key is '\??\C:\Users\IEUser\ntuser.dat' with a long GUID subkey and subkeys including 'AppEvents', 'Console', 'Control Panel', 'Environment', 'EUDC', 'Keyboard Layout', 'Network', 'Printers', 'Software', 'System', and 'Volatile Environment'. The third key is '\??\C:\System Volume Information\Syscache.hve' with a GUID subkey and a 'DefaultObjectStore' subkey. The terminal has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The top of the window shows a taskbar with various application icons and system status icons on the right.

```
Session Actions Edit View Help

Registry: \REGISTRY\MACHINE\BCD00000000
Key name: NewStoreRoot (S)
Last updated: 2019-01-06 18:02:44 UTC+0000

Subkeys:
(S) Description
(S) Objects

Values:

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: CMI-CreateHive{6A1C4018-979D-4291-A7DC-7AED1C75B67C} (S)
Last updated: 2019-01-06 15:02:48 UTC+0000

Subkeys:
(S) AppEvents
(S) Console
(S) Control Panel
(S) Environment
(S) EUDC
(S) Keyboard Layout
(S) Network
(S) Printers
(S) Software
(S) System
(V) Volatile Environment

Values:

Registry: \??\C:\System Volume Information\Syscache.hve
Key name: {a5b8a9ac-608c-11e5-a266-8af578093dd2} (S)
Last updated: 2019-01-06 15:02:50 UTC+0000

Subkeys:
(S) DefaultObjectStore
```

```

default@kali: ~/mem_lab
Session Actions Edit View Help

(default@kali)-[~/mem_lab]
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.registry.printkey --key "Volatile Environment"
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile
- 0x87c10370 Key [NONAME]\Volatile Environment - - -
- 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\Volatile Environment - - -
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\Volatile Environment - - -
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\Volatile Environment - - -
- 0x8b46d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\Volatile Environment - - -
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Volatile Environment 1 N/A True
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 REG_SZ \??\C:\Users\IEUser\ntuser.dat\Volatile Environment LOGONSERVER \\IE8WIN7 True
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 REG_SZ \??\C:\Users\IEUser\ntuser.dat\Volatile Environment USERDOMAIN IE8WIN7 True
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 REG_SZ \??\C:\Users\IEUser\ntuser.dat\Volatile Environment USERNAME IEUser True
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 REG_SZ \??\C:\Users\IEUser\ntuser.dat\Volatile Environment USERPROFILE C:\Users\IEUser True
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 REG_SZ \??\C:\Users\IEUser\ntuser.dat\Volatile Environment HOMEPATH \Users\IEUser True
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 REG_SZ \??\C:\Users\IEUser\ntuser.dat\Volatile Environment HOMEDRIVE C: True
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 REG_SZ \??\C:\Users\IEUser\ntuser.dat\Volatile Environment APPDATA C:\Users\IEUser\AppData\Roaming True
2019-01-06 15:02:48.000000 UTC 0x8b5d39b0 REG_SZ \??\C:\Users\IEUser\ntuser.dat\Volatile Environment LOCALAPPDATA C:\Users\IEUser\AppData\Local True
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\Volatile Environment -
- -
- 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE\Volatile Environment -

```

```

(default@kali)~[~/mem_lab]
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.registry.printkey --key "Microso
ft\Windows NT\CurrentVersion\ProfileList"
Volatility 3 Framework 2.28.2
Progress: 100.00
PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile
- 0x87c10370 Key [ONAME]\Microsoft\Windows NT\CurrentVersion\ProfileList -
- 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\Microsoft\Windows NT\CurrentVersion
\ProfileList -
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\Microsoft\Windows NT\CurrentVersi
on\ProfileList -
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\Mic
rosoft\Windows NT\CurrentVersion\ProfileList -
- 0x8b46d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\Micro
soft\Windows NT\CurrentVersion\ProfileList -
- 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Microsoft\Windows NT\CurrentV
ersion\ProfileList -
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\Microsoft\Windows NT\Cu
rrentVersion\ProfileList -
2009-07-14 04:41:12.000000 UTC 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE
\Microsoft\Windows NT\CurrentVersion\ProfileList S-1-5-18 N/A False
2015-09-21 19:14:17.000000 UTC 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE
\Microsoft\Windows NT\CurrentVersion\ProfileList S-1-5-19 N/A False
2015-09-21 19:14:17.000000 UTC 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE

```

```

default@kali: ~/mem_lab

Session Actions Edit View Help
-      0x8b46d008      Key      \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002
-      0x8b5d39b0      Key      \??\C:\Users\IEUser\ntuser.dat\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002
-      0x97104008      Key      \SystemRoot\System32\Config\SECURITY\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002
-
2015-09-21 09:51:49.000000 UTC 0x9711d9c8      REG_EXPAND_SZ  \SystemRoot\System32\Config\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002 ProfileImagePath C:\Users\sshd_server False
2015-09-21 09:51:49.000000 UTC 0x9711d9c8      REG_DWORD      \SystemRoot\System32\Config\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002 Flags 0 False
2015-09-21 09:51:49.000000 UTC 0x9711d9c8      REG_DWORD      \SystemRoot\System32\Config\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002 State 256 False
2015-09-21 09:51:49.000000 UTC 0x9711d9c8      REG_BINARY      \SystemRoot\System32\Config\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002 Sid
01 05 00 00 00 00 00 05 15 00 00 00 af 07 56 66 .....Vf
7e ca 36 36 e6 f5 33 46 ea 03 00 00 ~.66..3F.... False
2015-09-21 09:51:49.000000 UTC 0x9711d9c8      REG_DWORD      \SystemRoot\System32\Config\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002 ProfileLoadTimeLow 0 False
2015-09-21 09:51:49.000000 UTC 0x9711d9c8      REG_DWORD      \SystemRoot\System32\Config\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002 ProfileLoadTimeHigh 0 False
2015-09-21 09:51:49.000000 UTC 0x9711d9c8      REG_DWORD      \SystemRoot\System32\Config\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002 RefCount 1 False
-      0x981f9008      Key      \SystemRoot\System32\Config\DEFAULT\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002
-
-      0x98209518      Key      \SystemRoot\System32\Config\SAM\Microsoft\Windows NT\Curr

```

```

default@kali: ~/mem_lab

Session Actions Edit View Help

ion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1000 - - -
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\Microsoft\Windows NT\CurrentVe
rsion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1000 - - -
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\
Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-
1000 - - -
- 0x8b46d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\Mi
crosoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-10
00 - - -
- 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Microsoft\Windows NT\Curre
ntVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1000 - - -
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\Microsoft\Windows NT
\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1000 - - -
-
2019-01-06 15:07:15.000000 UTC 0x9711d9c8 REG_EXPAND_SZ \SystemRoot\System32\Conf
ig\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446
-1177810406-1000 ProfileImagePath C:\Users\IEUser False
2019-01-06 15:07:15.000000 UTC 0x9711d9c8 REG_DWORD \SystemRoot\System32\Conf
ig\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446
-1177810406-1000 Flags 0 False
2019-01-06 15:07:15.000000 UTC 0x9711d9c8 REG_DWORD \SystemRoot\System32\Conf
ig\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446
-1177810406-1000 State 0 False
2019-01-06 15:07:15.000000 UTC 0x9711d9c8 REG_BINARY \SystemRoot\System32\Conf
ig\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446
-1177810406-1000 Sid
01 05 00 00 00 00 00 05 15 00 00 00 af 07 56 66 .....Vf
7e ca 36 36 e6 f5 33 46 e8 03 00 00 ~.66..3F.... False
2019-01-06 15:07:15.000000 UTC 0x9711d9c8 REG_DWORD \SystemRoot\System32\Conf
ig\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446
-1177810406-1000 ProfileLoadTimeLow 0 False
2019-01-06 15:07:15.000000 UTC 0x9711d9c8 REG_DWORD \SystemRoot\System32\Conf
ig\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446
-1177810406-1000 ProfileLoadTimeHigh 0 False
2019-01-06 15:07:15.000000 UTC 0x9711d9c8 REG_DWORD \SystemRoot\System32\Conf

```

```

default@kali: ~/mem_lab

Session Actions Edit View Help

ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      ReportBootOk      1      False
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      Shell      explorer.exe      False
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      PreCreateKnownFolders {A520A1A4
-1780-4FF6-BD18-167343C5AF16} False
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      Userinit      C:\Windows\system
32\userinit.exe, False
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      VMApplet      SystemPropertiesP
erformance.exe /pagefile False
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      WinStationsDisabled 0      F
alse
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_DWORD \SystemRoot\System32\Conf
ig\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon      DisableCAD      1      F
alse
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      scremoveoption 0      False
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_DWORD \SystemRoot\System32\Conf
ig\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon      ShutdownFlags 214748368
7      False
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      DefaultDomainName      F
alse
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      DefaultUserName IEUser False
2019-01-06 15:02:47.000000 UTC 0x9711d9c8 REG_SZ \SystemRoot\System32\Config\SOFTW
ARE\Microsoft\Windows NT\CurrentVersion\Winlogon      AutoAdminLogon 1      False
-      0x981f9008      Key      \SystemRoot\System32\Config\DEFAULT\Microsoft\Windows NT\
CurrentVersion\Winlogon -      -
-      0x98209518      Key      \SystemRoot\System32\Config\SAM\Microsoft\Windows NT\Curr
entVersion\Winlogon -      -
-      0x982d55c0      Key      \REGISTRY\MACHINE\BCD00000000\Microsoft\Windows NT\Curren

```

```

default@kali: ~/mem_lab
Session Actions Edit View Help

(default@kali)-[~/mem_lab]
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.registry.hivelist
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Offset FileFullPath File output
0x87c10370 Disabled
0x87c1c008 \REGISTRY\MACHINE\SYSTEM Disabled
0x87c459c8 \REGISTRY\MACHINE\HARDWARE Disabled
0x889c0430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT Disabled
0x8b46d008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT Disabled
0x8b5d39b0 \??\C:\Users\IEUser\ntuser.dat Disabled
0x97104008 \SystemRoot\System32\Config\SECURITY Disabled
0x9711d9c8 \SystemRoot\System32\Config\SOFTWARE Disabled
0x981f9008 \SystemRoot\System32\Config\DEFAULT Disabled
0x98209518 \SystemRoot\System32\Config\SAM Disabled
0x982d55c0 \REGISTRY\MACHINE\BCD00000000 Disabled
0xa620b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat D
isabled
0xa65269c8 \??\C:\Users\sshd_server\ntuser.dat Disabled
0xa6539260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat D
isabled
0xa7d2a9c8 \??\C:\System Volume Information\Syscache.hve Disabled
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7fcc4a2435
e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in __del__
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable

(default@kali)-[~/mem_lab]
$

```

```

(default@kali)-[~/mem_lab]
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.registry.printkey --offset 0x
87c459c8 --key "DESCRIPTION\System\CentralProcessor\0"
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_BINARY \REGISTRY\MACHINE\HARDWAR
E\DESCRIPTION\System\CentralProcessor\0 Component Information
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ..... False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_SZ \REGISTRY\MACHINE\HARDWARE\DESCRI
PTION\System\CentralProcessor\0 Identifier x86 Family 6 Model 158 Stepping 9 F
alse
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_FULL_RESOURCE_DESCRIPTOR \REGISTRY
\MACHINE\HARDWARE\DESCRIPTION\System\CentralProcessor\0 Configuration Data
ff ff ff ff ff ff ff ff 00 00 00 00 00 00 00 00 ..... False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_SZ \REGISTRY\MACHINE\HARDWARE\DESCRI
PTION\System\CentralProcessor\0 ProcessorNameString Intel(R) Core(TM) i7-7700 CPU @ 3
.60GHz False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_SZ \REGISTRY\MACHINE\HARDWARE\DESCRI
PTION\System\CentralProcessor\0 VendorIdentifier GenuineIntel False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_DWORD \REGISTRY\MACHINE\HARDWAR
E\DESCRIPTION\System\CentralProcessor\0 FeatureSet 2697805823 False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_DWORD \REGISTRY\MACHINE\HARDWAR
E\DESCRIPTION\System\CentralProcessor\0 ~MHz 3600 False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_DWORD \REGISTRY\MACHINE\HARDWAR
E\DESCRIPTION\System\CentralProcessor\0 Update Status 2 False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 REG_BINARY \REGISTRY\MACHINE\HARDWAR
E\DESCRIPTION\System\CentralProcessor\0 Update Signature

```

```

default@kali: ~/mem_lab
Session Actions Edit View Help

(default@kali)-[~/mem_lab]
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.registry.printkey --key "DESCRIPTION\System"
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile
- 0x87c10370 Key [NONAME]\DESCRIPTION\System - - -
- 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\DESCRIPTION\System - - -
-
2019-01-06 18:02:40.000000 UTC 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\DESCRI
PTION\System CentralProcessor N/A False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\DESCRI
PTION\System FloatingPointProcessor N/A False
2019-01-06 18:02:40.000000 UTC 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\DESCRI
PTION\System MultifunctionAdapter N/A False
2019-01-06 18:02:44.000000 UTC 0x87c459c8 REG_BINARY \REGISTRY\MACHINE\HARDWAR
E\DESCRIPTION\System Component Information
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ..... False
2019-01-06 18:02:44.000000 UTC 0x87c459c8 REG_SZ \REGISTRY\MACHINE\HARDWARE\DESCRI
PTION\System Identifier AT/AT COMPATIBLE False
2019-01-06 18:02:44.000000 UTC 0x87c459c8 REG_FULL_RESOURCE_DESCRIPTOR \REGISTRY
\MACHINE\HARDWARE\DESCRIPTION\System Configuration Data
ff ff ff ff ff ff ff 00 00 00 00 02 00 00 00 .....
05 00 00 00 0c 00 00 00 00 00 00 00 00 00 00 00 .....
80 00 ff 03 00 00 3f 00 fe 00 01 00 05 00 00 00 .....?.....
08 00 00 00 00 00 00 00 00 00 00 00 00 00 0c 00 .....
00 00 04 00 .... False
2019-01-06 18:02:44.000000 UTC 0x87c459c8 REG_SZ \REGISTRY\MACHINE\HARDWARE\DESCRI
PTION\System SystemBiosDate 06/23/99 False
2019-01-06 18:02:44.000000 UTC 0x87c459c8 REG_MULTI_SZ \REGISTRY\MACHINE\HARDWAR
E\DESCRIPTION\System SystemBiosVersion VBox - 1
False
2019-01-06 18:02:44.000000 UTC 0x87c459c8 REG_DWORD \REGISTRY\MACHINE\HARDWAR

```

```

False
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\
DESCRIPTION\System -
- 0x8b46d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\DE
SCRIPTION\System -
- 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\DESCRIPTION\System -
- -
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\DESCRIPTION\System -
- -
- 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE\DESCRIPTION\System -
- -
- 0x981f9008 Key \SystemRoot\System32\Config\DEFAULT\DESCRIPTION\System -
- -
- 0x98209518 Key \SystemRoot\System32\Config\SAM\DESCRIPTION\System -
- -
- 0x982d55c0 Key \REGISTRY\MACHINE\BCD00000000\DESCRIPTION\System -
- -
- 0xa620b008 Key \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrCl
ass.dat\DESCRIPTION\System -
- 0xa65269c8 Key \??\C:\Users\sshd_server\ntuser.dat\DESCRIPTION\System -
- -
- 0xa6539260 Key \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\
UsrClass.dat\DESCRIPTION\System -
- 0xa7d2a9c8 Key \??\C:\System Volume Information\Syscache.hve\DESCRIPTION
\System -
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7fb177c575
e0>:

```

```

default@kali: ~/mem_lab
Session Actions Edit View Help
Set001\Enum"
Volatility 3 Framework 2.28.2
Progress: 100.00
Last Write Time Hive Offset Type Key Name Data Volatile
PDB scanning finished

- 0x87c10370 Key [NONAME]\ControlSet001\Enum - - -
2018-09-17 17:50:45.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum ACPI N/A False
2009-07-14 04:39:14.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum DISPLAY N/A False
2009-07-14 04:33:39.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum HTREE N/A False
2015-09-22 05:26:23.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum IDE N/A False
2015-09-22 05:22:57.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum LPTENUM N/A False
2019-01-06 18:02:40.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum PCI N/A False
2009-07-14 04:33:39.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum PCIIDE N/A False
2019-01-06 15:07:50.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum Root N/A False
2009-07-14 04:33:44.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum SCSI N/A False
2015-09-21 09:41:14.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum STORAGE N/A False
2009-07-14 04:54:03.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum SW N/A False
2019-01-06 15:02:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum USBSTOR N/A False
2015-09-21 18:14:52.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum VMBUS N/A False
2019-01-06 15:03:07.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0
01\Enum WpdBusEnumRoot N/A False
2019-01-06 15:03:07.000000 UTC 0x87c1c008 REG_DWORD \REGISTRY\MACHINE\SYSTEM\Con
trolSet001\Enum NextParentID.daba3ff.2 2 False

```

```

default@kali: ~/mem_lab
Session Actions Edit View Help

(default@kali)-[~/mem_lab]
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.registry.printkey --key "ControlSet001\Enum\PCI"
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile

- 0x87c10370 Key [NONAME]\ControlSet001\Enum\PCI - - -
2009-07-14 04:55:53.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_10008DEV_00546SUBSYS_1F0910286REV_01 N/A False
2009-07-14 04:55:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_10026DEV_515E6SUBSYS_01E610286REV_02 N/A False
2015-09-22 07:23:31.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_10228DEV_20006SUBSYS_200010226REV_40 N/A False
2015-09-22 05:22:32.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_106B8DEV_003F6SUBSYS_000000006REV_00 N/A False
2009-07-14 04:55:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_14E48DEV_16596SUBSYS_01E610286REV_11 N/A False
2015-09-22 02:40:27.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_15AD6DEV_07906SUBSYS_079015AD6REV_02 N/A False
2015-09-22 02:40:27.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_15AD6DEV_07A06SUBSYS_07A015AD6REV_01 N/A False
2009-07-14 04:55:55.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_80868DEV_032C6SUBSYS_000000006REV_09 N/A False
2019-01-06 17:36:24.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_80868DEV_100E6SUBSYS_001E80866REV_02 N/A False
2015-09-22 05:22:32.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_80868DEV_12376SUBSYS_000000006REV_02 N/A False
2019-01-06 18:02:40.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_80868DEV_265C6SUBSYS_000000006REV_00 N/A False
2009-07-14 04:55:53.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_80868DEV_27786SUBSYS_01E610286REV_00 N/A False
2009-07-14 04:55:53.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\PCI VEN_80868DEV_27796SUBSYS_01E610286REV_00 N/A False
2009-07-14 04:55:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet0

```

```

default@kali: ~/mem_lab
Session Actions Edit View Help
Last Write Time Hive Offset Type Key Name Data Volatile
- 0x87c10370 Key [ONAME]\CurrentControlSet - - -
2019-01-06 18:02:40.000000 UTC 0x87c1c008 REG_LINK \REGISTRY\MACHINE\SYSTEM\Cur
rentControlSet SymbolicLinkValue \Registry\Machine\System\ControlSet001 True
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\CurrentControlSet - - -
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\Cur
rentControlSet - - -
- 0x8b46d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\Curre
ntControlSet - - -
- 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\CurrentControlSet - -
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\CurrentControlSet - -
- 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE\CurrentControlSet - -
- 0x981f9008 Key \SystemRoot\System32\Config\DEFAULT\CurrentControlSet - -
- 0x98209518 Key \SystemRoot\System32\Config\SAM\CurrentControlSet - -
- 0x982d55c0 Key \REGISTRY\MACHINE\BCD00000000\CurrentControlSet - - -
- 0xa620b008 Key \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass
.dat\CurrentControlSet - - -
- 0xa65269c8 Key \??\C:\Users\sshd_server\ntuser.dat\CurrentControlSet - -
- 0xa6539260 Key \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\Usr
Class.dat\CurrentControlSet - - -
- 0xa7d2a9c8 Key \??\C:\System Volume Information\Syscache.hve\CurrentControl
Set - - -
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7f167484f5e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in __del__
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable

```

```

default@kali: ~/mem_lab
Session Actions Edit View Help
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.registry.printkey --key "ControlSet001\Control\ComputerName\ComputerName"
Volatility 3 Framework 2.28.2
Progress: 100.00
PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile
- 0x87c10370 Key [ONAME]\ControlSet001\Control\ComputerName\ComputerName - - -
2015-09-21 09:48:51.000000 UTC 0x87c1c008 REG_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\ComputerName\ComputerName (Default) mmsrvc False
2015-09-21 09:48:51.000000 UTC 0x87c1c008 REG_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\ComputerName\ComputerName ComputerName IE8WIN7 False
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\ControlSet001\Control\ComputerName\ComputerName - - -
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\ControlSet001\Control\ComputerName\ComputerName - - -
- 0x8b46d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\ControlSet001\Control\ComputerName\ComputerName - - -
- 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\ControlSet001\Control\ComputerName\ComputerName - - -
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\ControlSet001\Control\ComputerName\ComputerName - - -
- 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE\ControlSet001\Control\ComputerName\ComputerName - - -
- 0x981f9008 Key \SystemRoot\System32\Config\DEFAULT\ControlSet001\Control\ComputerName\ComputerName - - -
- 0x98209518 Key \SystemRoot\System32\Config\SAM\ControlSet001\Control\ComputerName\ComputerName - - -
- 0x982d55c0 Key \REGISTRY\MACHINE\BCD00000000\ControlSet001\Control\ComputerName\ComputerName - - -
- 0xa620b008 Key \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Control\ComputerName\ComputerName - - -
- 0xa65269c8 Key \??\C:\Users\sshd_server\ntuser.dat\ControlSet001\Control\ComputerName\ComputerName - - -
- 0xa6539260 Key \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Control\ComputerName\ComputerName - - -
- 0xa7d2a9c8 Key \??\C:\System Volume Information\Syscache.hve\ControlSet001\Control\ComputerName\ComputerName - - -
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7f540154b5e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in __del__
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable

```

```

Session Actions Edit View Help
[default@kali:~/mem_lab]
[default@kali]~/mem_lab
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.netscan | grep TCPv4
0*21478bb0 100.0 TCPv4 0.0.0.0 49155 0.0.0.0 0 LISTENING 432 services.exe N/A
0*269ff708 TCPv4 0.0.0.0 49156 0.0.0.0 0 LISTENING 440 lsass.exe N/A
0*3e043b10 TCPv4 192.168.56.8 49177 192.168.56.5 80 CLOSED - -
0*3e446008 TCPv4 0.0.0.0 49154 0.0.0.0 0 LISTENING 880 svchost.exe N/A
0*3e446098 TCPv4 0.0.0.0 49154 0.0.0.0 0 LISTENING 880 svchost.exe N/A
0*3e46e0c0 TCPv4 192.168.56.8 139 0.0.0.0 0 LISTENING 4 System N/A
0*3e671988 TCPv4 0.0.0.0 445 0.0.0.0 0 LISTENING 4 System N/A
0*3e72e530 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 672 svchost.exe N/A
0*3e72ee48 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 672 svchost.exe N/A
0*3e736388 TCPv4 0.0.0.0 49152 0.0.0.0 0 LISTENING 340 wininit.exe N/A
0*3e736d98 TCPv4 0.0.0.0 49152 0.0.0.0 0 LISTENING 340 wininit.exe N/A
0*3e79d528 TCPv4 0.0.0.0 49153 0.0.0.0 0 LISTENING 724 svchost.exe N/A
0*3e7a1618 TCPv4 0.0.0.0 49153 0.0.0.0 0 LISTENING 724 svchost.exe N/A
0*3e8f0620 TCPv4 0.0.0.0 49156 0.0.0.0 0 LISTENING 440 lsass.exe N/A
0*3e9b3e50 TCPv4 0.0.0.0 49155 0.0.0.0 0 LISTENING 432 services.exe N/A
0*3ee1aa88 TCPv4 0.0.0.0 22 0.0.0.0 0 LISTENING 2016 sshd.exe N/A
0*3ee1bc18 TCPv4 0.0.0.0 22 0.0.0.0 0 LISTENING 2016 sshd.exe N/A
0*3fcba008 TCPv4 192.168.56.8 49178 192.168.56.5 80 CLOSED - -
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7f088594b5e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in __del__
    File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable
[default@kali]~/mem_lab
$

```

```

[default@kali]~/mem_lab
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.pslist --pid 4
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
4 0 System 0x8413a908 73 518 N/A False 2019-01-06 18:02:44.000000 UTC N/A Disabled
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7f8a1c2575e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in __del__
    File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable
[default@kali]~/mem_lab
$

```

```

Session Actions Edit View Help
[default@kali:~/mem_lab]
[default@kali]~/mem_lab
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem yarascan.YaraScan --yara-string "192.168.56.5" > process_1.txt
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7fd2138575e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in __del__
    File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable
[default@kali]~/mem_lab
$ ls -l process_1.txt
-rw-rw-r-- 1 default default 60 Sep 22 06:36 process_1.txt
[default@kali]~/mem_lab
$

```

```

[default@kali]~/mem_lab
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.pslist --pid 3988
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
3988 348 conhost.exe 0x842ddd28 2 52 1 False 2019-01-06 15:06:11.000000 UTC N/A Disabled
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7f79788575e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in __del__
    File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable
[default@kali]~/mem_lab
$

```

```

Session Actions Edit View Help
[default@kali:~/mem_lab]
python3 ~/Volatility3/vol.py -f memdumpwin7.mem windows.dumpfiles --pid 3988
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Cache FileObject FileName Result
DataSectionObject 0x84351808 StaticCache.dat Error dumping file
SharedCacheMap 0x84351808 StaticCache.dat file.0x84351808.0x859ad508.SharedCacheMap.StaticCache.dat.vacb
ImageSectionObject 0x8591c920 cryptbase.dll file.0x8591c920.0x852af4c8.ImageSectionObject.cryptbase.dll
ImageSectionObject 0x8582ab20 user32.dll file.0x8582ab20.0x8580c3b0.ImageSectionObject.user32.dll
ImageSectionObject 0x8594e4e0 uxtheme.dll file.0x8594e4e0.0x859645d8.ImageSectionObject.uxtheme.dll
DataSectionObject 0x859454c0 StaticCache.dat Error dumping file
SharedCacheMap 0x859454c0 StaticCache.dat file.0x859454c0.0x859ad508.SharedCacheMap.StaticCache.dat.vacb
ImageSectionObject 0x85972708 dwmapi.dll file.0x85972708.0x859725e0.ImageSectionObject.dwmapi.dll
ImageSectionObject 0x858b9950 cryptbase.dll file.0x858b9950.0x858b9828.ImageSectionObject.cryptbase.dll
ImageSectionObject 0x85950d58 comctl32.dll file.0x85950d58.0x8594b460.ImageSectionObject.comctl32.dll
ImageSectionObject 0x85828df0 KernelBase.dll file.0x85828df0.0x8597db0d.ImageSectionObject.KernelBase.dll
ImageSectionObject 0x852b4560 rpcrt4.dll file.0x852b4560.0x85829008.ImageSectionObject.rpcrt4.dll
ImageSectionObject 0x858343b0 sechost.dll file.0x858343b0.0x858284e8.ImageSectionObject.sechost.dll
ImageSectionObject 0x85828460 lpk.dll file.0x85828460.0x858298d0.ImageSectionObject.lpk.dll
ImageSectionObject 0x85834468 shlwapi.dll file.0x85834468.0x85829130.ImageSectionObject.shlwapi.dll
ImageSectionObject 0x85827458 kernel32.dll file.0x85827458.0x85834700.ImageSectionObject.kernel32.dll
ImageSectionObject 0x852ba10 oleaut32.dll file.0x852ba10.0x85829740.ImageSectionObject.oleaut32.dll
ImageSectionObject 0x8582a038 msctf.dll file.0x8582a038.0x85831438.ImageSectionObject.msctf.dll
ImageSectionObject 0x94fa8d0 ntdll.dll file.0x94fa8d0.0x94df118.ImageSectionObject.ntdll.dll
ImageSectionObject 0x85181da0 msvcrt.dll file.0x85181da0.0x85181300.ImageSectionObject.msvcrt.dll
ImageSectionObject 0x8582de88 gdi32.dll file.0x8582de88.0x8580a308.ImageSectionObject.gdi32.dll
DataSectionObject 0x85181428 advapi32.dll file.0x85181428.0x858cd58.DataSectionObject.advapi32.dll
ImageSectionObject 0x858272e0 advapi32.dll file.0x858272e0.0x858272e0.ImageSectionObject.advapi32.dll
ImageSectionObject 0x852b3038 ole32.dll file.0x852b3038.0x8582d148.ImageSectionObject.ole32.dll
ImageSectionObject 0x851ebc98 apisetschema.dll file.0x851ebc98.0x877db998.ImageSectionObject.apisetschema.dll
DataSectionObject 0x8582b2e0 imm32.dll file.0x8582b2e0.0x852b5c10.DataSectionObject.imm32.dll
ImageSectionObject 0x8582b2e0 imm32.dll file.0x8582b2e0.0x8582ce48.ImageSectionObject.imm32.dll
Exception ignored while calling deallocator <function MagicDetect._del_ at 0x7f48ced4b5e0>:
Traceback (most recent call last):

```

```

Session Actions Edit View Help
[default@kali:~/mem_lab]
python3 ~/Volatility3/vol.py -f memdumpwin7.mem windows.registry.printkey --key "ControlSet001\Control\DeviceClasses"
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile
- 0x87c10370 Key [ONAME]\ControlSet001\Control\DeviceClasses - - -
2019-01-06 18:14:52.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {0eb06031-5213-4934-818b-38d90ced39db} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {180a7b1b-b511-4ae5-8318-a65c837b6661} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {2acfc660-1301-11d2-b082-00aa0c91ef8b} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {2e3d6580-5819-42ca-84ae-d38003bae505} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {32412632-86c8-44a2-9b5c-50d1417354f5} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {3375baf4-9e15-4b30-b765-67acb1d6d87b} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {34d14be3-dee4-41c8-9ae7-6b174977c192} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {35fa2e29-ea23-4236-96ae-3afebacba440} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {3abf67d2-71c4-462a-8a92-1e6861e6a727} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {4116f60b-25b3-4662-b732-99a6b11edec07} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {4afa3d53-74a7-11d0-be5e-00aa0c906287} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {4d1e55b2-1f6f-11cf-88cb-001110000030} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {4d36e978-e325-11ce-bf1c-08002be10318} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {53f56307-b6bf-11d0-94f2-00aa0c91ef8b} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {53f56308-b6bf-11d0-94f2-00aa0c91ef8b} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {53f56309-b6bf-11d0-94f2-00aa0c91ef8b} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {53f5630a-b6bf-11d0-94f2-00aa0c91ef8b} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {53f56311-b6bf-11d0-94f2-00aa0c91ef8b} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {57164f39-9115-4e78-ab55-382f3bd5422d} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {5b45201d-f2f2-4f3b-85bb-30ff1f953599} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {6ac27878-a6fa-4155-ba85-f98f491d4f33} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {72631e54-78a4-11d0-bc7f-00aa00b0732a} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {8665f9b5-3f07-4e97-b7df-24c5d0a8cc8b} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {952f6e30-d0ae-497b-adca-e08a0b175cf1} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {97f76ef8-f883-11d0-af1f-0000f800845c} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {a9a0f4e7-5a45-4096-b027-8a841e8c03e6} N/A False
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {e849804e-c719-43d8-ac88-96b894c191e2} N/A False
2019-01-06 15:03:08.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses {f33fcd04-d1ac-4e8e-9a30-19bbd4b108ae} N/A False

```

```

Session Actions Edit View Help
File /usr/lib/python3/dist-packages/magic/compat.py, line 126, in close
TypeError: 'NoneType' object is not callable
[default@kali:~/mem_lab]
python3 ~/Volatility3/vol.py -f memdumpwin7.mem windows.registry.printkey --key "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b}"
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile
- 0x87c10370 Key [ONAME]\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
2019-01-06 18:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} ##?#IDE#Disk#BOX_HARDDISK_
2019-01-06 15:17:53.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} ##?#IDE#Disk#Virtual_HD_
2019-01-06 15:02:41.000000 UTC 1.1.0 \53f56307-b6bf-11d0-94f2-00aa0c91ef8b N/A False ##?#IDE#Disk#Wmware_Virtual_ID
E_Hard_Drive 00000001562eb498060.0.0\53f56307-b6bf-11d0-94f2-00aa0c91ef8b N/A False ##?#SCSI#Disk#Ven_Dell6Prod_V
2019-01-06 15:02:41.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} ##?#USB#STOR#Disk#Ven_General#
Prod_Disk#Rev_5.006651becf48606.60\53f56307-b6bf-11d0-94f2-00aa0c91ef8b N/A False
- 0x87c45928 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0x8846d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0x885d39b0 Key \??\C:\Users\IEUser\ntuser.dat\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0x97110408 Key \SystemRoot\System32\Config\SOFTWARE\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0x861f9808 Key \SystemRoot\System32\Config\DEFAULT\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0x98209518 Key \SystemRoot\System32\Config\SAM\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0x982d55c0 Key \REGISTRY\MACHINE\BCD\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0xa620b008 Key \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0xa65269c8 Key \??\C:\Users\sshd_server\ntuser.dat\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0xa6539260 Key \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
- 0xa7d2a9c8 Key \??\C:\System Volume Information\Syscache.hve\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00aa0c91ef8b} - - -
Exception ignored while calling deallocator <function MagicDetect._del_ at 0x7fc619427e0>:
Traceback (most recent call last):

```

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```

Session Actions Edit View Help

--(default@kali) ~/mem_lab
$ python3 ~/volatility3/vol.py -f memdumpwin7.mem windows.registry.printkey --key "ControlSet001\Enum\USBSTOR"
Volatility 3 Framework 2.28.2
Progress: 100.00
Last Write Time Hive Offset PDB scanning finished
Type Key Name Data Volatile

- 0x87c10370 Key [ONAME] ControlSet001\Enum\USBSTOR
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR DiskVen_General6Prod_UDisk6Rev_5.00 N/A False
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\ControlSet001\Enum\USBSTOR
- 0x889c0430 Key \??(C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\ControlSet001\Enum\USBSTOR
- 0x8b46d008 Key \??(C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\ControlSet001\Enum\USBSTOR
- 0x8b5d39b0 Key \??(C:\Users\IEUser\ntuser.dat\ControlSet001\Enum\USBSTOR
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\ControlSet001\Enum\USBSTOR
- 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE\ControlSet001\Enum\USBSTOR
- 0x9b1f9008 Key \SystemRoot\System32\Config\DEFAULT\ControlSet001\Enum\USBSTOR
- 0x9b209518 Key \SystemRoot\System32\Config\SAM\ControlSet001\Enum\USBSTOR
- 0x9b2d55c0 Key \REGISTRY\MACHINE\BCD00000000\ControlSet001\Enum\USBSTOR
- 0xa620b008 Key \??(C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Enum\USBSTOR
- 0xa65269c8 Key \??(C:\Users\sshd_server\ntuser.dat\ControlSet001\Enum\USBSTOR
- 0xa6539260 Key \??(C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Enum\USBSTOR
- 0xa7d2a9c8 Key \??(C:\System Volume Information\Syscache.hve\ControlSet001\Enum\USBSTOR
Exception ignored while calling deallocator <function MagicDetect._del_ at 0x7f706f85f5e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in _del_
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable

--(default@kali) ~/mem_lab
$

```

```

Session Actions Edit View Help

- 0xa620b008 Key \??(C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Enum\USBSTOR
- 0xa65269c8 Key \??(C:\Users\sshd_server\ntuser.dat\ControlSet001\Enum\USBSTOR
- 0xa6539260 Key \??(C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Enum\USBSTOR
- 0xa7d2a9c8 Key \??(C:\System Volume Information\Syscache.hve\ControlSet001\Enum\USBSTOR
Exception ignored while calling deallocator <function MagicDetect._del_ at 0x7f706f85f5e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in _del_
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable

--(default@kali) ~/mem_lab
$ python3 ~/volatility3/vol.py -f memdumpwin7.mem windows.registry.printkey --key "ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00"
Volatility 3 Framework 2.28.2
Progress: 100.00
Last Write Time Hive Offset PDB scanning finished
Type Key Name Data Volatile

- 0x87c10370 Key [ONAME] ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00 661bec0f48606_60 N/A False
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0x889c0430 Key \??(C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0x8b46d008 Key \??(C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0x8b5d39b0 Key \??(C:\Users\IEUser\ntuser.dat\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0x9711d9c8 Key \SystemRoot\System32\Config\SOFTWARE\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0x9b1f9008 Key \SystemRoot\System32\Config\DEFAULT\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0x9b209518 Key \SystemRoot\System32\Config\SAM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0x9b2d55c0 Key \REGISTRY\MACHINE\BCD00000000\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0xa620b008 Key \??(C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0xa65269c8 Key \??(C:\Users\sshd_server\ntuser.dat\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0xa6539260 Key \??(C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
- 0xa7d2a9c8 Key \??(C:\System Volume Information\Syscache.hve\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00
Exception ignored while calling deallocator <function MagicDetect._del_ at 0x7f395453f5e0>:
Traceback (most recent call last):
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 299, in _del_
  File "/usr/lib/python3/dist-packages/magic/compat.py", line 126, in close
TypeError: 'NoneType' object is not callable

```

```

Session Actions Edit View Help

- 0x87c10370 Key [ONAME] ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 Device Parameters N
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 LogConf N/A False
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 Properties N/A F
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 Control N/A True
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 DeviceDesc @disk
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_DWORD \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 Capabilities0
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_MULTI_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 HardwareID U
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_MULTI_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 CompatibleIDs
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 ContainerID {421e
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_DWORD \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 ConfigFlags 0
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 ClassGUID {4d36
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 Class DiskDrive F
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 Mfg @disk.inf,Xge
- 2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_SZ \REGISTRY\MACHINE\SYSTEM\ControlSet001\Enum\USBSTOR\DiskVen_General6Prod_UDisk6Rev_5.00\661bec0f48606_60 Service disk False

```

```

Session Actions Edit View Help
23 00 36 00 26 00 31 00 62 00 65 00 63 00 30 00 #.6.6.1.b.e.c.0.
66 00 34 00 38 00 26 00 30 00 26 00 5f 00 26 00 f.4.8.6.0.8.8.
30 00 23 00 7b 00 35 00 33 00 66 00 35 00 36 00 0.#.[.5.3.f.5.6.
33 00 30 00 37 00 2d 00 62 00 36 00 62 00 66 00 3.0.7.-.b.6.b.f.
2d 00 31 00 31 00 64 00 30 00 2d 00 39 00 34 00 -.1.1.d.0.-.9.4.
66 00 32 00 2d 00 30 00 30 00 61 00 30 00 63 00 00 f.2.-.0.0.a.0.c.
39 00 31 00 65 00 66 00 62 00 38 00 62 00 7d 00 9.1.e.f.b.8.b.].
False
2019-01-06 15:02:54.000000 UTC 0x87c1c008 REG_BINARY \REGISTRY\MACHINE\SYSTEM\MountedDevices \DosDevices\F:
5f 00 3f 00 3f 00 5f 00 55 00 53 00 42 00 53 00 .??.U.S.B.S.
54 00 4f 00 52 00 23 00 44 00 69 00 73 00 6b 00 T.O.R.#.D.i.s.k.
26 00 56 00 65 00 6e 00 5f 00 47 00 65 00 6e 00 6.V.e.n._.G.e.n.
65 00 72 00 61 00 6c 00 26 00 50 00 72 00 6f 00 e.r.a.l.6.P.r.o.
64 00 5f 00 55 00 44 00 69 00 73 00 6b 00 26 00 d..U.D.i.s.k.6.
52 00 65 00 76 00 5f 00 35 00 2e 00 30 00 30 00 R.e.v..5..0.0.
23 00 36 00 26 00 31 00 62 00 65 00 63 00 30 00 #.6.6.1.b.e.c.0.
66 00 34 00 38 00 26 00 30 00 26 00 5f 00 26 00 f.4.8.6.0.8.8.
30 00 23 00 7b 00 35 00 33 00 66 00 35 00 36 00 0.#.[.5.3.f.5.6.
33 00 30 00 37 00 2d 00 62 00 36 00 62 00 66 00 3.0.7.-.b.6.b.f.
2d 00 31 00 31 00 64 00 30 00 2d 00 39 00 34 00 -.1.1.d.0.-.9.4.
66 00 32 00 2d 00 30 00 30 00 61 00 30 00 63 00 00 f.2.-.0.0.a.0.c.
39 00 31 00 65 00 66 00 62 00 38 00 62 00 7d 00 9.1.e.f.b.8.b.].
False
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\MountedDevices - - -
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\MountedDevices - - -
- 0x8b46d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\MountedDevices - - -
- 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\MountedDevices - - -
- 0x97104008 Key \SystemRoot\System32\Config\SECURITY\MountedDevices - - -
- 0x971109e8 Key \SystemRoot\System32\Config\SOFTWARE\MountedDevices - - -
- 0x981f9008 Key \SystemRoot\System32\Config\DEFAULT\MountedDevices - - -
- 0x98209518 Key \SystemRoot\System32\Config\SAM\MountedDevices - - -
- 0x982d55c0 Key \REGISTRY\MACHINE\BCD00000000\MountedDevices - - -
- 0xa620b008 Key \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat\MountedDevices - - -
- 0xa65269c8 Key \??\C:\Users\sshd_server\ntuser.dat\MountedDevices - - -
- 0xa6539260 Key \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat\MountedDevices - - -
- 0xa7d2e5e8 Key \??\C:\System Volume Information\Syscache.hve\MountedDevices - - -
Exception ignored while calling deallocator <function MagicDetect.__del__ at 0x7f30b764b5e0>:
Traceback (most recent call last):

```

```

Session Actions Edit View Help
(default@kali)-[~/mem_lab]
$ python3 ~/volatility3/vol.py -f memdumpWin7.mem windows.registry.printkey --key "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"
Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Last Write Time Hive Offset Type Key Name Data Volatile
- 0x87c10370 Key [NONAME]\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 - - -
- 0x87c1c008 Key \REGISTRY\MACHINE\SYSTEM\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 - - -
- 0x87c459c8 Key \REGISTRY\MACHINE\HARDWARE\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 - - -
- 0x889c0430 Key \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 - - -
- 0x8b46d008 Key \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 - - -
2015-09-21 09:19:50.000000 UTC 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 CPC N/A False
2019-01-06 15:03:06.000000 UTC 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 F N/A False
2019-01-06 15:03:07.000000 UTC 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 {421e7e52-11dd-11e9-b3cf-08002709e15d} N/A False
2019-01-06 14:32:25.000000 UTC 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 {762f4ebc-60ea-11e5-83af-806e6f6e6963} N/A False
2015-09-21 22:21:32.000000 UTC 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 {8358fe6d-60fa-11e5-bb4a-806e6f6e6963} N/A False
2015-09-21 09:44:39.000000 UTC 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 {a5b8a980-608c-11e5-a266-806e6f6e6963} N/A False
2015-09-21 09:44:39.000000 UTC 0x8b5d39b0 Key \??\C:\Users\IEUser\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 {a5b8a983-608c-11e5-a266-806e6f6e6963} N/A False

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